

ASSIGNMENT 7:

Sequential Agentic Workflows using LangGraph and Gemini 2.5 Flash

INTRODUCTION

Modern enterprise AI systems increasingly rely on intelligent workflows capable of executing multiple tasks sequentially while maintaining context and decision continuity. Traditional automation systems often struggle with handling dynamic reasoning, contextual understanding, and adaptive decision-making. To overcome these limitations, organizations are adopting agentic AI workflows that combine Large Language Models (LLMs), workflow orchestration frameworks, state management, and tool integration.

This assignment focuses on building sequential agentic workflows using LangGraph and Gemini 2.5 Flash. The workflows simulate real-world enterprise processes where each stage depends on the output generated by the previous stage. The assignment demonstrates how enterprise systems can automate complex decision-making processes while maintaining structured execution and contextual memory.

The implementation covers two practical enterprise use cases:

1. Telecom SIM Activation and Fraud Verification Workflow
2. Healthcare Appointment Prioritization Workflow

Both workflows use sequential node execution, state management, tool calling, and LLM-based reasoning to automate business decisions.

OBJECTIVE

The primary objective of this assignment is to understand and implement sequential AI workflows using LangGraph. The assignment aims to demonstrate:

- Sequential execution of AI workflow nodes
- State management across workflow stages
- Integration of tools inside AI workflows
- Gemini-based reasoning and intelligent classification
- Enterprise-grade workflow orchestration
- Multi-step decision-making pipelines

TECHNOLOGIES USED

The following technologies were used in this assignment:

1. Python

Python was used as the core programming language for implementing the workflows and business logic.

2. LangGraph

LangGraph was used to create stateful sequential workflows. It enables node-based orchestration where outputs from one node are passed to the next node.

3. LangChain

LangChain was used for integrating Gemini models into the workflows.

4. Gemini 2.5 Flash

Gemini 2.5 Flash was used as the Large Language Model for performing intelligent reasoning, classification, and decision-making.

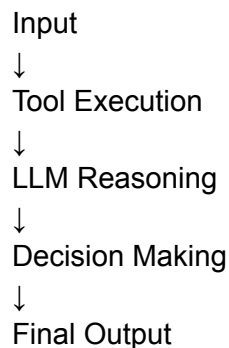
5. TypedDict

TypedDict was used for workflow state management to maintain structured data throughout workflow execution.

Sequential Agentic Workflow Concept

A sequential agentic workflow is a process where multiple AI agents or nodes execute tasks one after another. Each node performs a dedicated operation and passes its output to the next node.

General Workflow Structure:



The key characteristics of sequential workflows include:

- Step-by-step execution
- Dependency between nodes
- Persistent state management
- Intelligent decision-making
- Dynamic business logic processing

USE CASE 1 – Telecom SIM Activation and Fraud Verification Workflow

Business Problem

Telecom companies receive thousands of SIM activation requests daily. Manual verification of customer identity and fraud analysis is time-consuming and error-prone. To automate this process, an AI workflow is designed to:

- Verify customer KYC documents
- Detect fraud risks
- Analyze previous SIM request history
- Make final activation decisions

Workflow Architecture

The telecom workflow consists of three sequential stages:

1. KYC Verification Tool
2. Fraud Detection using Gemini
3. Final SIM Activation Decision

Step 1 – KYC Verification Tool

The first node validates customer identity documents such as Aadhaar and PAN cards.

Input Parameters:

- Aadhaar Status
- PAN Status

Business Logic:

- If both documents are valid → VERIFIED
- If any document is pending → PENDING_VERIFICATION
- Otherwise → INVALID_DOCUMENT

Expected Outputs:

- VERIFIED
- INVALID_DOCUMENT
- PENDING_VERIFICATION

Purpose:

This stage ensures that customer identity documents are validated before further processing.

Step 2 – Fraud Detection using Gemini

The second node uses Gemini 2.5 Flash for fraud risk analysis.

Input Parameters:

- Customer location
- Number of previous SIM requests
- KYC verification result

Gemini analyzes the provided information and classifies the customer into one of the following categories:

- LOW_RISK
- MEDIUM_RISK
- HIGH_RISK

Purpose:

This stage enables intelligent fraud detection using AI reasoning rather than fixed rules.

Step 3 – Final Activation Decision

The final node makes the activation decision based on the fraud risk classification.

Business Logic:

- LOW_RISK → SIM Activated
- MEDIUM_RISK → Manual Review Required
- HIGH_RISK → Application Rejected

Purpose:

This stage automates final business actions based on earlier workflow results.

Telecom Workflow State Management

The workflow state stores all intermediate and final outputs:

- Customer details
- KYC result
- Fraud risk classification
- Final activation decision

State management ensures smooth data transfer between nodes.

Telecom Workflow Execution Flow

Customer Input

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KYC Verification Tool

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Gemini Fraud Detection

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Final Activation Decision

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Output Generated

Sample Telecom Workflow Output

Customer Name: Maria Dominic

KYC Result: VERIFIED

Fraud Risk: LOW_RISK

Final Decision: SIM ACTIVATED

USE CASE 2 – Healthcare Appointment Prioritization Workflow

Business Problem

Hospitals receive patients with varying medical conditions and urgency levels. Manual prioritization can delay treatment for critical patients. To automate patient triage and consultation assignment, an AI-based workflow is developed.

The workflow performs:

- Symptom severity analysis
- Medical prioritization
- Consultation assignment

Workflow Architecture

The healthcare workflow contains three sequential stages:

1. Symptom Severity Tool
2. Gemini Medical Prioritization
3. Final Consultation Assignment

Step 1 – Symptom Severity Tool

The first node analyzes patient symptoms using predefined medical rules.

Input Parameters:

- Fever
- Oxygen level
- Heart rate
- Symptom duration

Business Logic:

- Oxygen below 90 or heart rate above 130 → CRITICAL
- Fever above 101 or symptom duration greater than 5 days → MODERATE
- Otherwise → STABLE

Expected Outputs:

- STABLE
- MODERATE
- CRITICAL

Purpose:

This stage performs initial patient condition assessment.

Step 2 – Gemini Medical Prioritization

The second node uses Gemini 2.5 Flash to determine consultation priority.

Input Parameters:

- Severity level
- Patient age
- Existing medical conditions

Gemini classifies patients into:

- EMERGENCY
- PRIORITY_CONSULTATION
- REGULAR_CONSULTATION

Purpose:

This stage performs intelligent medical prioritization using contextual reasoning.

Step 3 – Final Consultation Assignment

The final node assigns the appropriate medical consultation type.

Business Logic:

- EMERGENCY → ICU / Emergency Room
- PRIORITY_CONSULTATION → Specialist Doctor
- REGULAR_CONSULTATION → General Physician

Purpose:

This stage automates patient routing and consultation assignment.

Healthcare Workflow State Management

The workflow state stores:

- Patient information
- Severity level
- Consultation priority
- Final assignment

This allows sequential nodes to access previous outputs.

Healthcare Workflow Execution Flow

Patient Input

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Symptom Severity Tool

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Gemini Medical Prioritization

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Consultation Assignment

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Final Output

Sample Healthcare Workflow Output

Patient Name: Anita Verma

Severity Level: CRITICAL

Consultation Priority: EMERGENCY

Consultation Assignment: ICU / EMERGENCY ROOM

LangGraph Workflow Implementation

LangGraph was used to create directed sequential workflows.

The implementation involved:

- Creating workflow states using TypedDict
- Adding nodes for each workflow stage
- Connecting nodes using edges
- Compiling the workflow graph
- Executing the graph using invoke()

Sequential execution was implemented as:

Node 1 → Node 2 → Node 3

Each node updated the workflow state and passed it to the next node.

Role of Gemini 2.5 Flash

Gemini 2.5 Flash played a critical role in intelligent decision-making.

Functions performed by Gemini:

- Fraud risk classification
- Medical prioritization
- Contextual reasoning
- Dynamic classification

Role of Tool Calling

Tool calling was used for deterministic tasks.

Examples:

- KYC document verification
- Symptom severity calculation

Benefits of tool integration:

- Fast execution
- Rule-based validation
- Reliable business logic
- Structured outputs

State Management

State management is essential in sequential workflows because every node depends on outputs from earlier stages.

The workflow state stores:

- Inputs
- Intermediate outputs
- Final decisions

Benefits:

- Preserves workflow context
- Enables sequential execution
- Simplifies node communication
- Supports enterprise-scale workflows

Advantages of Sequential Agentic Workflows

- Modular architecture
- Better maintainability
- Intelligent automation

- Improved scalability
- Context-aware decision-making
- Enterprise process automation
- Reduced manual intervention

Conclusion

This assignment successfully demonstrated the implementation of sequential agentic workflows using LangGraph and Gemini 2.5 Flash. Two enterprise-level use cases were developed to showcase real-world workflow automation using AI-driven decision-making and sequential execution.

The telecom workflow automated SIM activation using KYC validation, fraud analysis, and final decision-making. The healthcare workflow automated patient prioritization using symptom analysis, AI reasoning, and consultation assignment.

The assignment highlighted the importance of combining tools, state management, and LLM reasoning to create scalable enterprise AI systems. These architectures represent the foundation of modern intelligent automation platforms used across industries such as telecom, healthcare, banking, insurance, and retail.